Date of Revision

CHICAGO MACHANICAL LUBRICATORS FOR VALVES AND CYLINDERS (Cont'd)

Testing

If lubricator is not delivering required amount of oil to any or all points, it should be tested for pressure and stroke as follows:

Disconnect oil lines at lubricator, screw a pressure gauge into the delivery opening. Hand-crank or ratchet driving arm until pressure over 310 pounds is registered on gauge. If pressure does not build up on the pressure stroke of the pump plunger when moved very slowly, it indicates a defective cylinder cup, or leak past pump rod seat. If pressure builds up on the pressure stroke but falls back rapidly after plunger has finished forcing stroke, it indicates defective check valves in base of lubricator, and these should be examined and repaired. With gauge showing over 200 pounds, the pressure should not drop back more than 15 pounds per minute.

Follow same procedure on all feeds.

Test for pump stroke the same as described in setting feeds.

If engine is equipped with terminal checks, these should be tested.

Inspection - Incoming

Examine to see if there is any water in bowl, by opening the 1/8" drain valve in base of lubricator. All water must be drained from bowl. Water may enter the bowl from a cracked heater, be poured in with oil when filling or back up through one or more oil delivery lines due to defective check valves. If more than a few teaspoonfuls of water is found in lubricator bowl, it should be reported so that check valves, (also terminal checks if so equipped), may be examined. Inspector for water should be made before lubricator is filled.

Check amount of oil remaining in bowl, to determine if lubricator has been delivering regular amount of oil. Any variation should be reported.

Check oil delivery pipes and connections for leaks and broken pipes.

Check pipe clamps and bracket and lubricator support bolts for loose or missing cotter pins, nuts or bolts.

Check driving mechanism for loose nuts, missing cotter pins or excessive wear. There must not be more than 1/8" total wear or lost motion in driving mechanism parts.

Hand-crank lubricator slowly several turns to see that resistance is offered on each pump stroke.

Inspection - Outgoing

See that lubricator has been filled. Inspect connections for indications of oil leaks. Driving motion parts must be in good condition, all nuts and cotter pins in place.

Filling cover and measuring rod, where so equipped, must be securely in place.

When the atmospheric temperature is 40 degrees F. or colder, or where locomotive will encounter this temperature during trip, steam heat must be supplied to lubricator. The steam heat to lubricator must be in operative condition before locomotive leaves terminal. To check the steam heat to lubricator, be sure steam is turned on at fountain, and steam valve at lubricator is open. Open drain valve at the choke fitting, until all water is drained out and steam appears, then close drain valve. In case lubricator cools off from lack of steam heat due to choke being stopped up, the drain valve may be opened sufficiently to allow a light blow of steam to show. This must be reported at next terminal so that repairs may be made. Unless choke fitting is stopped up the drain valve at choke fitting, should remain closed at all times while locomotive is in service. Steam heat to oil delivery lines should be on all the time engine is in service from October to May.

Hand-crank lubricator slowly several turns to see that resistance is offered on each pump stroke.

Valve oil will expand due to heat, but will not boil at temperature of steam from the boiler.

If oil is being forced out around filling cap after steam heat has been on for some time, lubricator may have been filled too full; heater chamber may have a crack in it, or there may be water in the lubricator.

Defects should be repaired before engine is placed in service.

Monthly Inspection

Test, with pressure gauge, each oil delivery, for pressure, to determine if cups, check valves in lubricator base or terminal checks are leaking. If leaking, make necessary repairs.

Quarterly Inspection

Remove base. Wash base and inside of lubricator bowl with hot water under pressure. Then wash base and inside of bowl with rag and kerosene, distillate, or gasoline. Do not use waste. Then blow clean and dry with air. Examine all parts and make necessary repairs. Reassemble lubricator, set feeds, check all oil delivery pipes and connections. (This includes terminal checks, if a part of the application.) See that oil is delivered at all points.

Semi-Annual Inspection - Same as Quarterly - General Overhauling

When locomotives receive general repairs, lubricators, piping and fittings at delivery end of oil pipes must be removed from engine. Oil delivery pipes must be annealed before being applied and blown out after being annealed, also after being reapplied. Lubricator and fittings and driving mechanism must be overhauled and put in first-class condition before being reapplied. Lubricator must be tested as outlined in quarterly test.

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